Linear Functions & Systems Maze **ANSWER KEY**

Name _____ Period _____

Start in the top-left room. Solve each problem. Go through the door that shows your answer. Show your work on the answer grid.

START

1 Write the equation of the line through $(-2, 7)$ and $(4, -5)$.

y = −2x + 3

2 Write, in slope-intercept form, the line through $(3, -4)$ perpendicular to $2x - 6y = 9$.

y = −3x + 5

3 Solve the system.

$$\begin{cases} 4x - 3y = 19 \\ 3x + 5y = 7 \end{cases}$$

(4, −1)

4 Which point is a solution of both inequalities?

$$\begin{cases} y > -x + 2 \\ y \le 2x - 1 \end{cases}$$

(3, 2)

$-\frac{4}{3}$

$y = 3x - 13$

$\frac{3}{4}$

4

150 min

(1, 3)

Find the slope of a line perpendicular to the line through $(-3, 1)$ and $(5, -5)$.

$-\frac{4}{3}$

6 Find the line of best fit (regression line).

x	1	2	3	4	5
y	52	58	61	69	75

$y = 5.7x + 63$

42

5 Maximize $P = 5x + 4y$ subject to $x \ge 0, y \ge 0, x + 2y \le 12, 3x + y \le 21$.

24

How many solutions does the system have?

$6x - 4y = 10$

$y = 1.5x - 2.5$

none

(−1, 2, 4)

y = 5.7x + 45.9

$y = 5.7x + 4.59$

46

35

exactly one

7 Solve the system.

$$\begin{cases} x + y + z = 5 \\ 2x - y + z = 9 \\ x + 2y - z = -4 \end{cases}$$

(2, 1, 4)

Boundary line and shading for $y \le -\frac{1}{2}x + 3$?

60 min

Plan A: \$20 plus \$0.10 per min. Plan B: \$5 plus \$0.25 per min. For how many minutes do they cost the same?

75 min

Find the x-intercept of $5x - 3y = 30$.

(2, −1, 4)

8

dashed, below

300

$3x - 2y = 7$

−6

10

8 A show sold 60 tickets for \$500: adult \$12, student \$8, child \$5. Adult tickets were twice the child tickets. How many student tickets?

48

9 A tank holds 340 gal at 5 min and 180 gal at 9 min, draining at a constant rate. How many gallons at 12 min?

60

10 Line through $(-1, 5)$ parallel to $3x + 2y = 8$. Write it in standard form.

3x + 2y = 7

FINISH

You made it!

Linear Functions Maze: Answer Grid **ANSWER KEY**

Name _____ Period _____

Write each answer in the order you found it, and show your work for that room.

1	Answer: **$y = -2x + 3$**	$m = \frac{-5 - 7}{4 - (-2)} = -2$; $7 = -2(-2) + b \to b = 3 \;\to\; y = -2x + 3$
2	Answer: **$y = -3x + 5$**	given slope $\frac{1}{3}$; perpendicular slope -3; $-4 = -3(3) + b \to b = 5 \;\to\; y = -3x + 5$
3	Answer: **$(4, -1)$**	5(1st) + 3(2nd): $29x = 116 \to x = 4$; then $y = -1 \;\to\; (4, -1)$
4	Answer: **$(3, 2)$**	$(3, 2)$: $2 > -3 + 2$ and $2 \le 2(3) - 1$ are both true; each other point fails one $\;\to\; (3, 2)$
5	Answer: **42**	vertices $(0, 0)$, $(7, 0)$, $(0, 6)$, $(6, 3)$ give $P = 0, 35, 24, 42 \;\to\; 42$
6	Answer: **$y = 5.7x + 45.9$**	mean $x = 3$, mean $y = 63$; slope $= \frac{57}{10} = 5.7$; $63 = 5.7(3) + b \to b = 45.9 \;\to\; y = 5.7x + 45.9$
7	Answer: **$(2, -1, 4)$**	eq. 1 + eq. 3: $2x + 3y = 1$; eq. 2 + eq. 3: $3x + y = 5$; so $x = 2$, $y = -1$, $z = 4 \;\to\; (2, -1, 4)$
8	Answer: **48**	$a = 2c$, $s = 60 - 3c$; $24c + 8(60 - 3c) + 5c = 500 \to c = 4$, $s = 48 \;\to\; 48$
9	Answer: **60**	rate $= \frac{180 - 340}{9 - 5} = -40$ gal per min; $180 + 3(-40) = 60 \;\to\; 60$
10	Answer: **$3x + 2y = 7$**	same slope, so $3x + 2y = 3(-1) + 2(5) = 7 \;\to\; 3x + 2y = 7$